

1.7 Classwork: Cumulative frequency graphs — Markscheme

Calculator allowed. Marks are not printed on the student sheet; they are recorded here. Total: 17 marks.

Throughout: a value read from a graph is accepted within the range given. Lines drawn on the graph that show the reading count as working for an M1. Where a later part uses an earlier reading, follow through (FT) from the student's value.

1. [Maximum mark: 9]

100 students; curve passes through (30, 15), (40, 25), (60, 50), (80, 75), (100, 90).

(a) Median at cumulative frequency 50: 60 minutes A1

Accept 59 to 61.

(b) $Q_1 = 40$ and $Q_3 = 80$ A1

$IQR = 80 - 40 = 40$ minutes A1

Accept Q_1 in 39 to 41 and Q_3 in 79 to 81. The second A1 is FT from the student's quartiles. Reading the quartiles at cumulative frequencies other than 25 and 75 earns A0A0.

(c) 15 students spent less than 30 minutes (M1)

$100 - 15 = 85$ students A1

Accept 84 to 86. The M1 is for reading the curve at $t = 30$, seen or implied by a line on the graph. "15" as the final answer earns M1A0: it is the number below 30 minutes, not at least 30.

(d) 90% of 100 = 90 students, so read the curve at cumulative frequency 90 (M1)

100 minutes A1

Accept 99 to 101. Reading at cumulative frequency 10 (the 10th percentile, about 24 minutes) earns M0A0.

(e) $Q_1 - 1.5 \times IQR = 40 - 60 = -20$ and $Q_3 + 1.5 \times IQR = 80 + 60 = 140$ M1

$5 > -20$ and $118 < 140$, so neither the shortest nor the longest time is an outlier R1

no outliers AG

M1 for at least one boundary correctly calculated (FT from (b)). The R1 needs both comparisons, with the minimum and the maximum. Checking only the longest time earns M1R0.

2. [Maximum mark: 8]

120 apples; curve passes through (120, 10), (140, 35), (160, 75), (170, 90), (180, 105), (200, 120).

(a) Median at cumulative frequency 60: 152.5 grams **A1**

Accept 151 to 154. Reading at cumulative frequency 60.5 is also accepted.

(b) 90 apples have a mass of 170 g or less **(M1)**

$120 - 90 = 30$ large apples **A1**

Accept 29 to 31. "90" as the final answer earns M1A0.

(c) $p = 75 - 35 = 40$ **A1**

$q = 105 - 75 = 30$ **A1**

Check: $10 + 25 + 40 + 30 + 15 = 120$. Cumulative frequencies read as the frequencies ($p = 75$, $q = 105$) earn A0A0.

(d) Mid-interval values 110, 130, 150, 170, 190 with frequencies 10, 25, 40, 30, 15 **(M1)**

$\bar{x} = \frac{1100 + 3250 + 6000 + 5100 + 2850}{120} = \frac{18\,300}{120} = 152.5$ grams **A1**

GDC: mid-interval values in L1, frequencies in L2, One-Variable Statistics with frequency list L2: $\bar{x} = 152.5$.

FT from the student's p and q . Using the class boundaries (120, 140, ...) instead of the mid-interval values earns M0A0. The mean of the five mid-interval values alone (150) earns M0A0.

(e) The actual masses are not known; each apple is given the mid-interval value of its class. **R1**

Accept "the data are grouped", "mid-interval values are used". "The graph is not exact" alone earns R0: the mean is calculated from the table, and it would be an estimate even with exact frequencies.